

## Cumulative frequency tables and histograms



1

a 26

b 6

c 18

2

| Score ( $x$ ) | Frequency ( $f$ ) | Cumulative frequency ( $cf$ ) |
|---------------|-------------------|-------------------------------|
| 18            | 10                | 10                            |
| 19            | 9                 | 19                            |
| 20            | 3                 | 22                            |
| 21            | 7                 | 29                            |
| Total:        | 29                |                               |

3

a

| Number of Sightings | Number of Locations ( $f$ ) | Cumulative frequency ( $cf$ ) |
|---------------------|-----------------------------|-------------------------------|
| 7                   | 4                           | 4                             |
| 8                   | 3                           | 7                             |
| 9                   | 5                           | 12                            |
| 10                  | 2                           | 14                            |
| 11                  | 5                           | 19                            |
| 12                  | 1                           | 20                            |

b 6

c 14

d 9

4

a 2

b 12

c

| Score<br>( $x$ ) | Frequency<br>( $f$ ) | Cumulative<br>frequency ( $cf$ ) |
|------------------|----------------------|----------------------------------|
| 2                | 1                    | 1                                |
| 3                | 2                    | 3                                |
| 4                | 5                    | 8                                |
| 5                | 5                    | 13                               |
| 6                | 5                    | 18                               |
| 7                | 9                    | 27                               |
| 8                | 7                    | 34                               |
| 9                | 5                    | 39                               |
| 10               | 8                    | 47                               |
| 11               | 1                    | 48                               |
| 12               | 2                    | 50                               |



d

7

e 11

f 18

5

| Score | Frequency | Cumulative frequency |
|-------|-----------|----------------------|
| 25    | 10        | 10                   |
| 26    | 16        | 26                   |
| 27    | 14        | 40                   |
| 28    | 20        | 60                   |
| 29    | 16        | 76                   |
| 30    | 10        | 86                   |

6

a 14

b 2

c 0

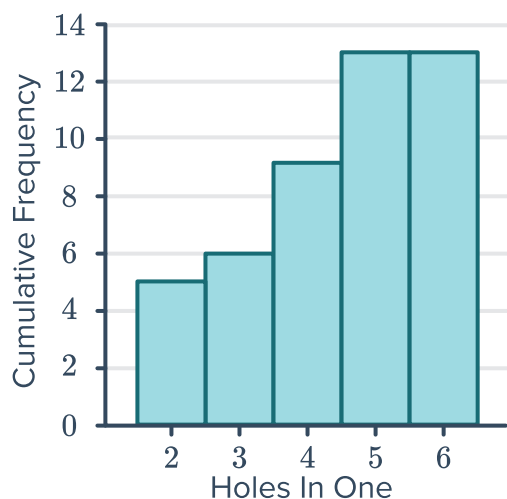
d 71.4%

7



a

b 45



c 6

8

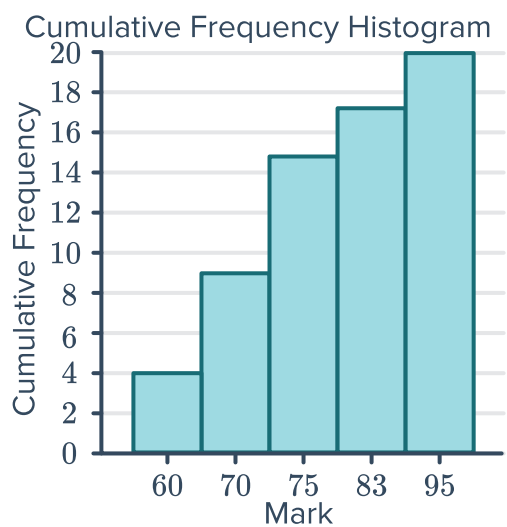
a The difference between the heights of two adjacent columns.

b 7

9

a

b 11



c 9

d 2

e 75

f 74.55

g 75

10



a

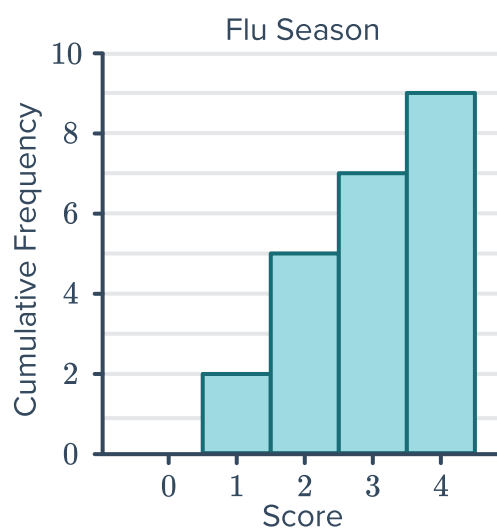
| Score ( $x$ ) | Frequency ( $f$ ) | $fx$ | Cumulative frequency |
|---------------|-------------------|------|----------------------|
| 0             | 0                 | 0    | 0                    |
| 1             | 2                 | 2    | 2                    |
| 2             | 3                 | 6    | 5                    |
| 3             | 2                 | 6    | 7                    |
| 4             | 2                 | 8    | 9                    |

b 9

c 2

d 2.44

e



f 4

## Cumulative frequency ogive

11

| Score | Frequency |
|-------|-----------|
| 84    | 2         |
| 85    | 2         |
| 86    | 4         |
| 87    | 6         |
| 88    | 2         |

12



13

a 30

b 10

c 10

d 20

e 30

f 20

14

a

i 12

ii 7

iii 17

b

i 675

ii 640

iii 760

c

i 2100

ii 1900

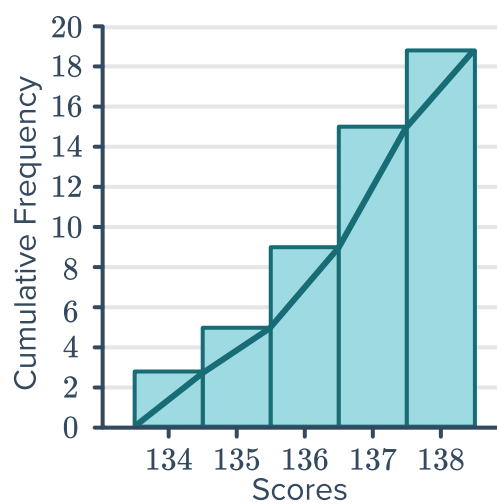
iii 2450

15

a

| Score ( $x$ ) | Frequency ( $f$ ) | Cumulative frequency ( $cf$ ) |
|---------------|-------------------|-------------------------------|
| 134           | 3                 | 3                             |
| 135           | 2                 | 5                             |
| 136           | 4                 | 9                             |
| 137           | 6                 | 15                            |
| 138           | 4                 | 19                            |
| Total:        | 19                |                               |

b



c 137

d 135



## Cumulative frequency and grouped data

16

a

i

| Class   | Class centre | Frequency | Cumulative frequency |
|---------|--------------|-----------|----------------------|
| 1 – 7   | 4            | 11        | 11                   |
| 8 – 14  | 11           | 14        | 25                   |
| 15 – 21 | 18           | 10        | 35                   |
| 22 – 28 | 25           | 15        | 50                   |
| 29 – 35 | 32           | 24        | 74                   |
| Totals  |              | 74        |                      |

ii 29 – 35

iii 22 – 28

iv 20.55

b

i

| Class   | Class centre | Frequency | Cumulative frequency |
|---------|--------------|-----------|----------------------|
| 1 – 9   | 5            | 8         | 8                    |
| 10 – 18 | 14           | 16        | 24                   |
| 19 – 27 | 23           | 4         | 28                   |
| 28 – 36 | 32           | 21        | 49                   |
| 37 – 45 | 41           | 16        | 65                   |
| Totals  |              | 65        |                      |

ii 28 – 36

iii 28 – 36

iv 25.91

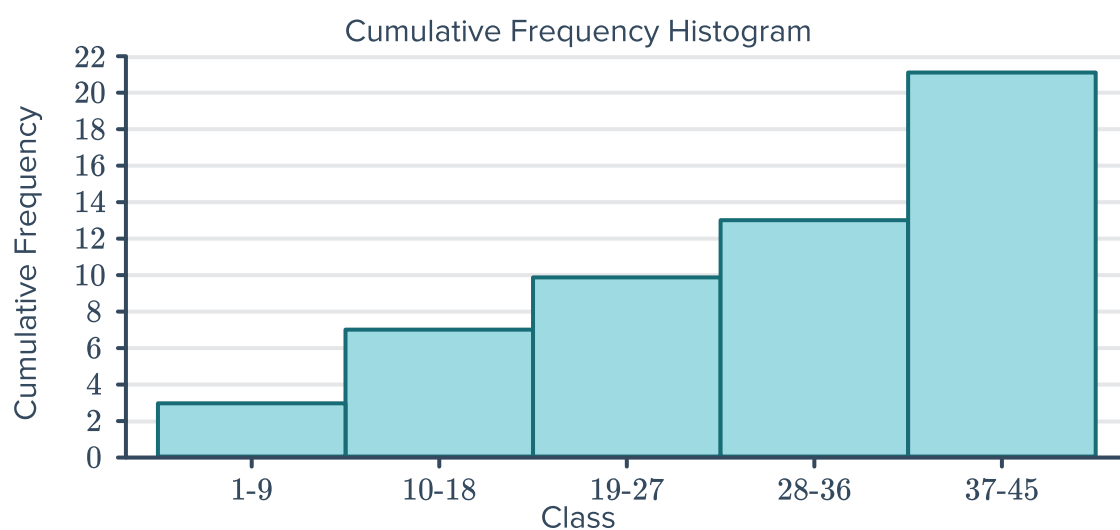
17



a

| Class   | Class centre | Frequency | Cumulative frequency |
|---------|--------------|-----------|----------------------|
| 1 – 9   | 5            | 3         | 3                    |
| 10 – 18 | 14           | 4         | 7                    |
| 19 – 27 | 23           | 3         | 10                   |
| 28 – 36 | 32           | 3         | 13                   |
| 37 – 45 | 41           | 8         | 21                   |
| Totals  |              | 21        |                      |

b



c 37 – 45

d 28 – 36

18

a

| Score ( $x$ ) | Frequency ( $f$ ) | Cumulative frequency ( $cf$ ) |
|---------------|-------------------|-------------------------------|
| 1 – 4         | 4                 | 4                             |
| 5 – 8         | 7                 | 11                            |
| 9 – 12        | 11                | 22                            |
| 13 – 16       | 7                 | 29                            |
| 17 – 20       | 4                 | 33                            |

b 33

c 4

d Symmetrical

e 12

19



a

| Score ( $x$ ) | Frequency ( $f$ ) | Cumulative frequency ( $cf$ ) |
|---------------|-------------------|-------------------------------|
| 1 – 5         | 7                 | 7                             |
| 6 – 10        | 15                | 22                            |
| 11 – 15       | 8                 | 30                            |
| 16 – 20       | 11                | 41                            |
| 21 – 25       | 2                 | 43                            |
| 26 – 30       | 1                 | 44                            |

b 44

c 5

d Positively skewed

e 10

20

a

| Score ( $x$ ) | Frequency ( $f$ ) | Cumulative frequency ( $cf$ ) |
|---------------|-------------------|-------------------------------|
| 20 – 24       | 9                 | 9                             |
| 25 – 29       | 18                | 27                            |
| 30 – 34       | 37                | 64                            |
| 35 – 39       | 14                | 78                            |
| 40 – 44       | 9                 | 87                            |
| 45 – 49       | 6                 | 93                            |
| 50 – 54       | 3                 | 96                            |

b 96

c 5

d Positively skewed

e 34



21



a

| Class interval | Class centre( $cc$ ) | Frequency( $f$ ) | $f \times cc$ | Cumul. Freq. |
|----------------|----------------------|------------------|---------------|--------------|
| 51 – 55        | 53                   | 1                | 53            | 1            |
| 56 – 60        | 58                   | 4                | 232           | 5            |
| 61 – 65        | 63                   | 2                | 126           | 7            |
| 66 – 70        | 68                   | 3                | 204           | 10           |
| 71 – 75        | 73                   | 2                | 146           | 12           |
| Total          |                      | 12               | 761           |              |

b 56 – 60

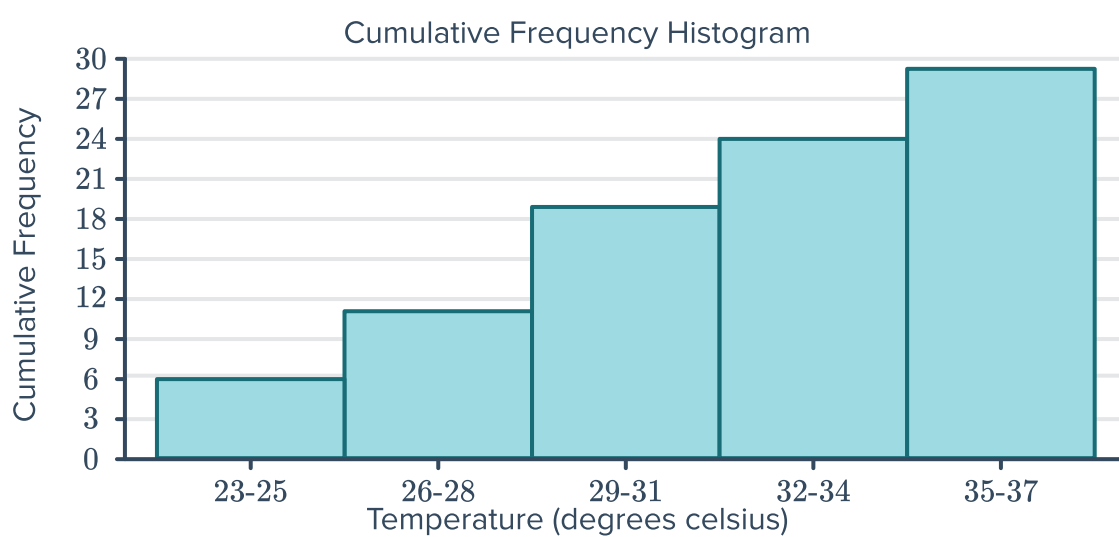
c 63.42

22

a

| Class   | Class centre | Frequency | Cumulative frequency |
|---------|--------------|-----------|----------------------|
| 23 – 25 | 24           | 6         | 6                    |
| 26 – 28 | 27           | 5         | 11                   |
| 29 – 31 | 30           | 8         | 19                   |
| 32 – 34 | 33           | 5         | 24                   |
| 35 – 37 | 36           | 5         | 29                   |
| Totals: |              | 29        |                      |

b



c 29 – 31 °C

d 29.79 °C

23



a

| Time Recorded     | Frequency | Cumulative frequency |
|-------------------|-----------|----------------------|
| 16 : 00 – 16 : 09 | 4         | 4                    |
| 16 : 10 – 16 : 19 | 2         | 6                    |
| 16 : 20 – 16 : 29 | 6         | 12                   |
| 16 : 30 – 16 : 39 | 7         | 19                   |
| 16 : 40 – 16 : 49 | 5         | 24                   |
| 16 : 50 – 16 : 59 | 1         | 25                   |

b 25

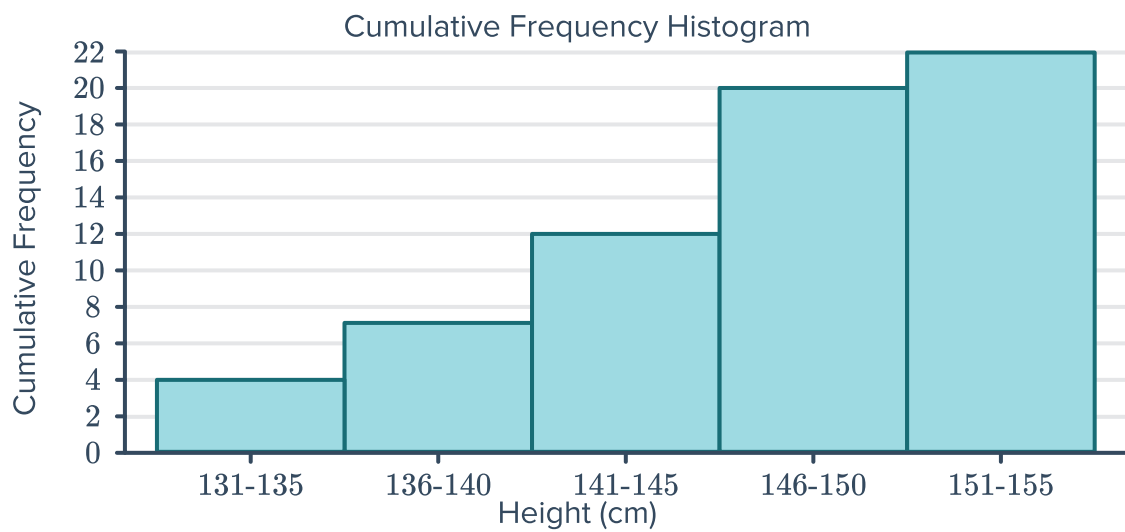
c 16:30 - 16:39

d 24

e 48%

24

a



b 2 students

c 20 students

d 5 students

e 146 – 150

f 148

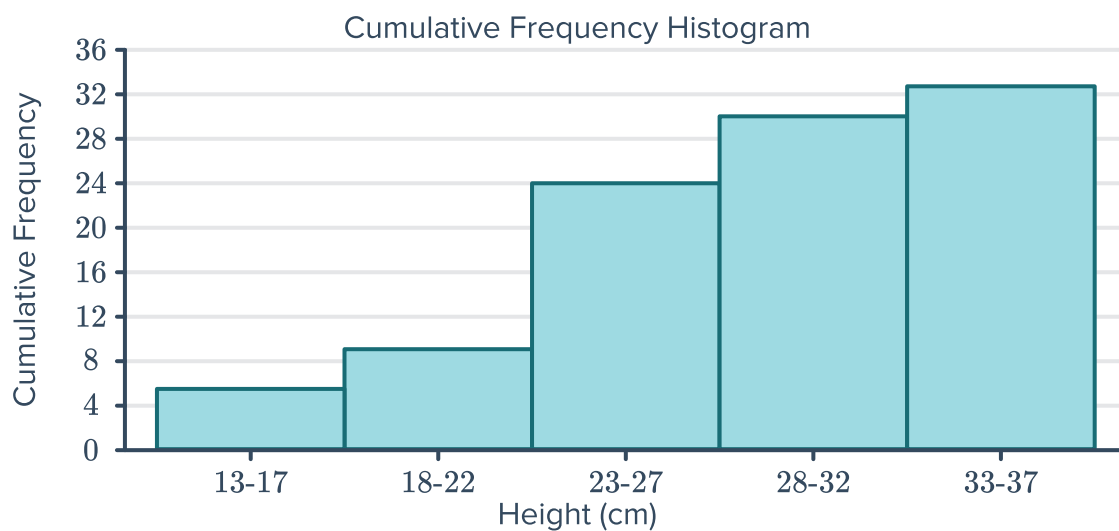
g 141 – 145

25



a 21 cm

b



c 24.55 cm

d 23 – 27

e 23 – 27

f 72.7%

g 27.3%

26

a

i 0.13 seconds

ii 0.132 seconds

b

| Reaction Time (seconds) | Class Centre | Frequency | Cumulative frequency |
|-------------------------|--------------|-----------|----------------------|
| $0.01 \leq t < 0.09$    | 0.05         | 61        | 61                   |
| $0.09 \leq t < 0.17$    | 0.13         | 70        | 131                  |
| $0.17 \leq t < 0.25$    | 0.21         | 69        | 200                  |

c

i 0.13 seconds

ii 0.133 seconds

d 0.758%